VECTOR BASICS AND GIVEN DIRECTIONS

1. Define a vector. Then write $\vec{A}$, $\vec{B}$ and $\vec{C}$ in magnitude [direction] form.

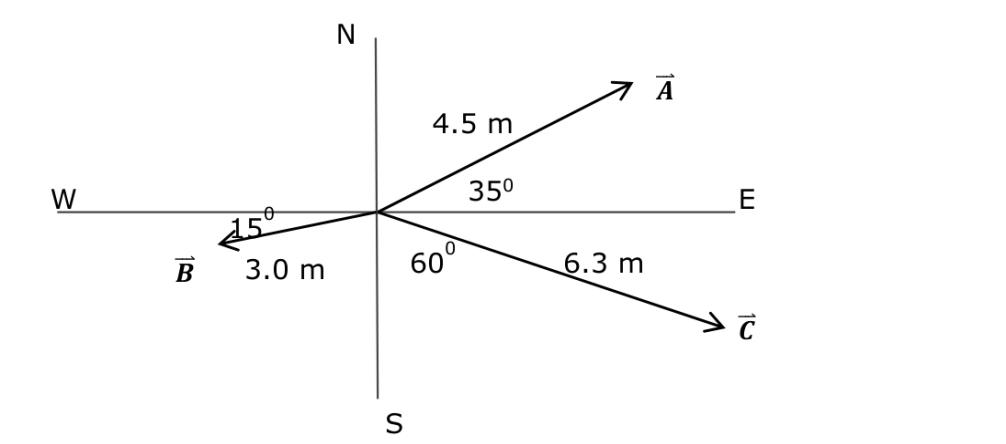
1 Understand what a vector contains

A vector has both a magnitude (size with units) and a direction. In the key, a symbol with an arrow is a vector; bars around it, such as $|\vec{A}|$, mean its scalar magnitude.

2 Read the angle from the first compass letter

For [N15°E], begin at north and turn 15° toward east. This is different from a true bearing, which is measured clockwise from north.

Original worksheet diagram (not to scale)



3 State each vector using its marked length and angle

$\vec{A}$ = 4.5 m [E35°N] • start east, then turn 35° north.

$\vec{B}$ = 3.0 m [W15°S] • start west, then turn 15° south.

$\vec{C}$ = 6.3 m [S60°E] • start south, then turn 60° east.

Equivalent direction for $\vec{C}$: [E30°S], because 90° - 60° = 30°. Either compass form describes the same arrow.

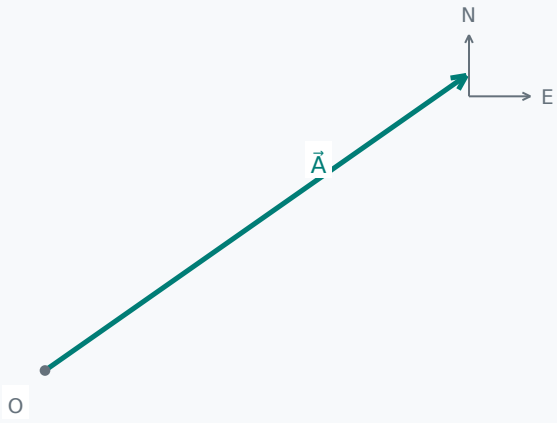
RESOLVE VECTOR $\vec{A}$ INTO COMPONENTS

1(b). Sketch the components of $\vec{A}$ = 4.5 m [E35°N]; find the magnitude of each component.

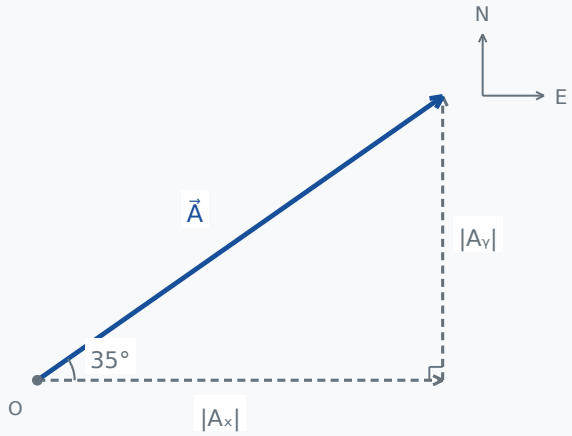
1 Identify the axis used to measure the given angle

The 35° angle starts at east. The horizontal x leg lies beside this angle and uses cosine; the vertical y leg lies across from it and uses sine.

Given vector and direction



Perpendicular component triangle



2 Calculate positive component magnitudes

$|A_x| = 4.5 \cos 35° \approx 3.686$ m

$|A_y| = 4.5 \sin 35° \approx 2.581$ m

Signed components for later addition: $A_x > 0$, $A_y > 0$. The question asks for magnitudes here, so both boxed lengths are positive.

$|A_x| = 3.7$ m $\quad |A_y| = 2.6$ m

For later operations, recalculate signed components from the original lengths and angles; retain calculator precision until the final result.

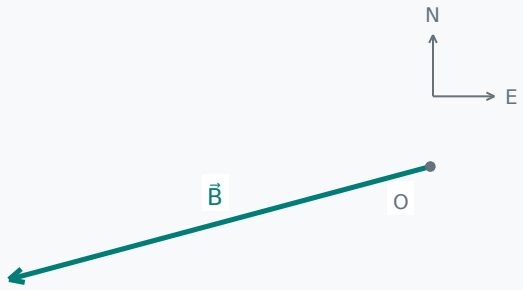
RESOLVE VECTOR $\vec{B}$ INTO COMPONENTS

1(b). Sketch the components of $\vec{B}$ = 3.0 m [W15°S]; find the magnitude of each component.

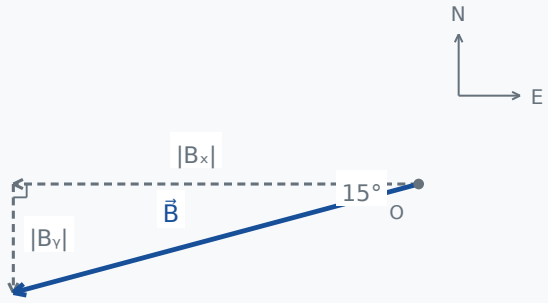
1 Identify the axis used to measure the given angle

The 15° angle starts at west. The horizontal x leg lies beside this angle and uses cosine; the vertical y leg lies across from it and uses sine.

Given vector and direction



Perpendicular component triangle



2 Calculate positive component magnitudes

$|B_x| = 3.0 \cos 15° \approx 2.898$ m

$|B_y| = 3.0 \sin 15° \approx 0.776$ m

Signed components for later addition: $B_x < 0$, $B_y < 0$. The question asks for magnitudes here, so both boxed lengths are positive.

$|B_x| = 2.9$ m $\quad |B_y| = 0.78$ m

For later operations, recalculate signed components from the original lengths and angles; retain calculator precision until the final result.

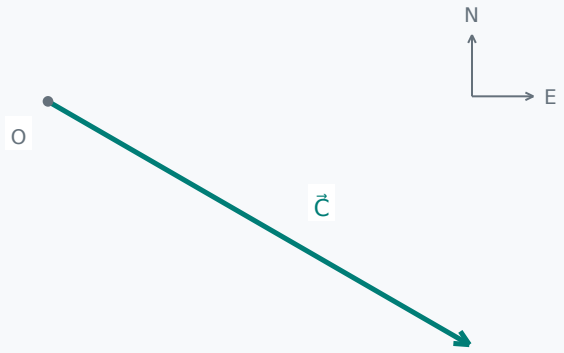
RESOLVE VECTOR $\vec{C}$ INTO COMPONENTS

1(b). Sketch the components of $\vec{C}$ = 6.3 m [S60°E]; find the magnitude of each component.

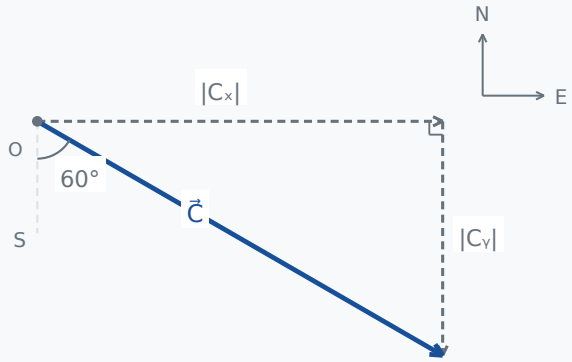
1 Identify the axis used to measure the given angle

The 60° angle starts at south. The vertical y leg lies beside this angle and uses cosine; the horizontal x leg lies across from it and uses sine.

Given vector and direction



Perpendicular component triangle



2 Calculate positive component magnitudes

$|C_x| = 6.3 \sin 60° \approx 5.456$ m

$|C_y| = 6.3 \cos 60° = 3.150$ m

Signed components for later addition: $C_x > 0$, $C_y < 0$. The question asks for magnitudes here, so both boxed lengths are positive.

$|C_x| = 5.5$ m $|C_y| = 3.2$ m

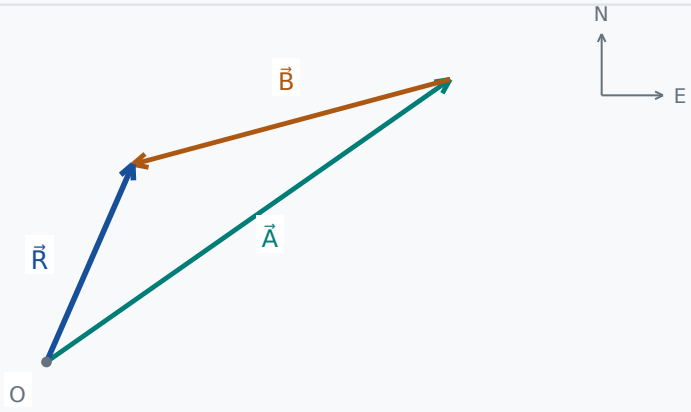
For later operations, recalculate signed components from the original lengths and angles; retain calculator precision until the final result.

DRAWING TWO-VECTOR SUMS

1. Draw $\vec{A} + \vec{B}$ and $\vec{B} + \vec{C}$ tip to tail. No calculations are required for these drawings.

1 Draw $\vec{A} + \vec{B}$

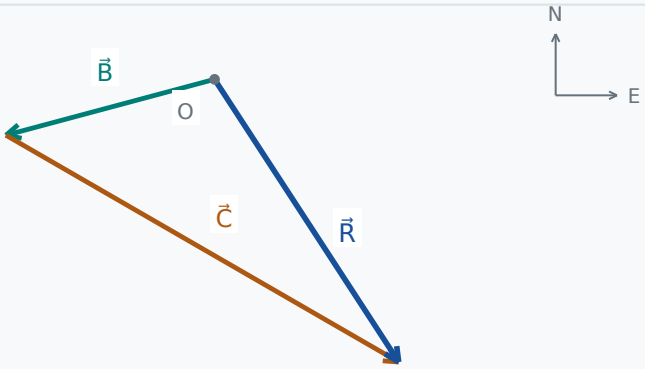
Original directions retained; arrows moved, not turned



Draw $\vec{A}$ first. Slide $\vec{B}$ until its tail touches $\vec{A}$'s tip. Draw the blue resultant from $\vec{A}$'s starting point to $\vec{B}$'s final tip.

2 Draw $\vec{B} + \vec{C}$

The order is $\vec{B}$, then $\vec{C}$



Keep $\vec{B}$ southwest and $\vec{C}$ southeast. Place $\vec{C}$'s tail on $\vec{B}$'s tip, then join the first tail to the final tip.

The blue arrow represents the net change in each drawing. $\vec{B} + \vec{C}$ is a graphical question here: no numerical $\vec{B} + \vec{C}$ answer is requested.

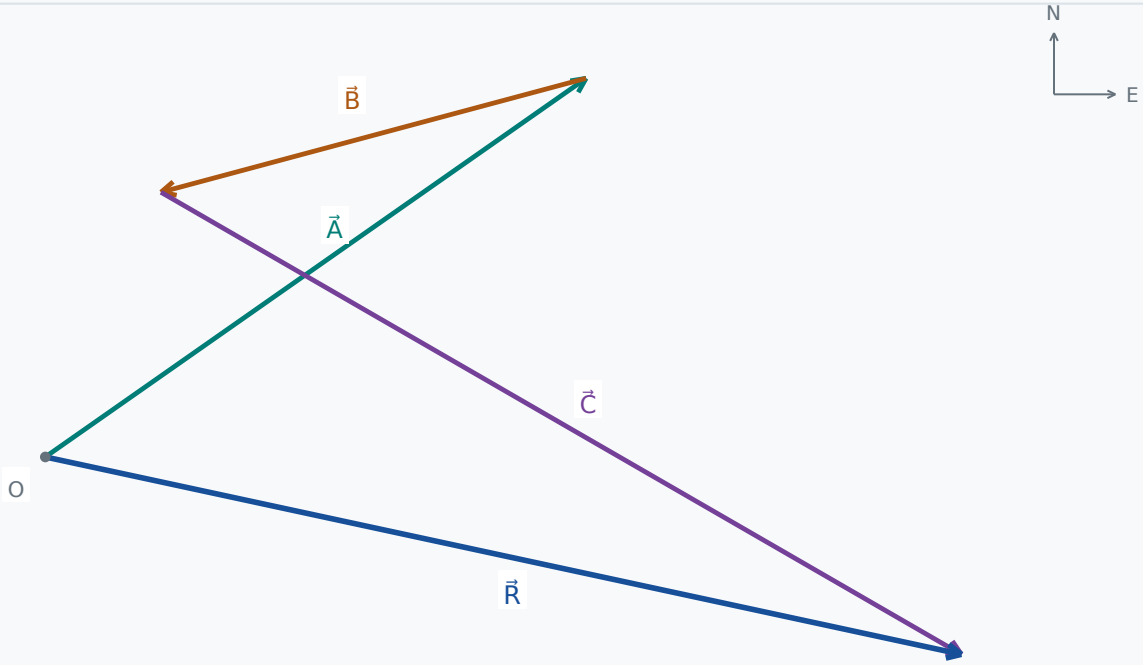
DRAWING $\vec{A}$ THREE-VECTOR SUM

1. Draw $\vec{A} + \vec{B} + \vec{C}$ tip to tail; no calculations are required for this part.

1 Place the three original arrows in order

Translate each arrow without changing its length or direction: $\vec{A}$ first, then $\vec{B}$ from $\vec{A}$'s tip, then $\vec{C}$ from $\vec{B}$'s tip.

Complete tip-to-tail path and resultant



2 Connect the first tail to the last tip

The blue arrow is the resultant $\vec{A} + \vec{B} + \vec{C}$. Its direction and length are found numerically in question 3, where components are requested.

Graphical result: one arrow from the first tail to the final tip

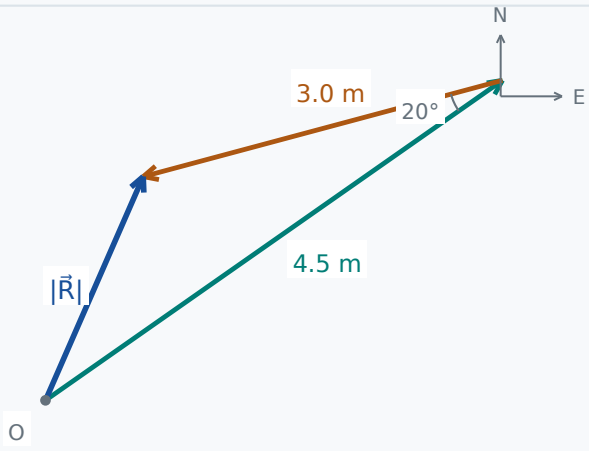
TRIANGLE METHOD FOR $\vec{A} + \vec{B}$

2. Use the sine and cosine laws to find the unknown side of $\vec{A} + \vec{B}$. First find the angle between the given vectors. Report the side length.

Original vectors from one origin



Tip-to-tail triangle



1 Determine the included triangle angle

$\vec{A}$ is [E35°N]; $\vec{B}$ is [W15°S]. Their original directions differ by 160°. The angle inside the tip-to-tail triangle is the supplement: 180° - 160° = 20°.

2 Apply the cosine law to the two known sides

$|\vec{R}|^2 = 4.5^2 + 3.0^2 - 2(4.5)(3.0)\cos 20°$

$|\vec{R}| = \sqrt{[4.5^2 + 3.0^2 - 2(4.5)(3.0)\cos 20°]} \approx 1.969340 \text{ m}$

3 Use the sine law as a triangle check

$\sin \varphi / 3.0 = \sin 20° / |\vec{R}|$

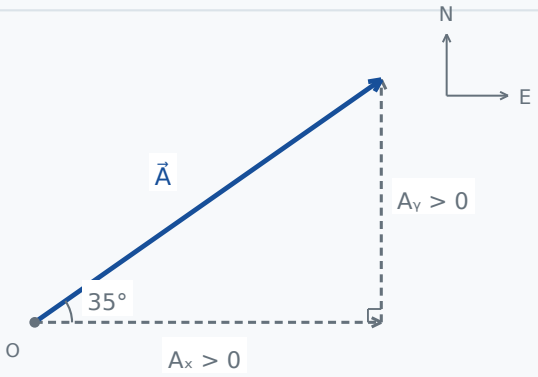
$\varphi = \sin^{-1}(3.0 \sin 20° / |\vec{R}|) \approx 31.401°$

Requested scalar: $|\vec{A} + \vec{B}| = 2.0$ m

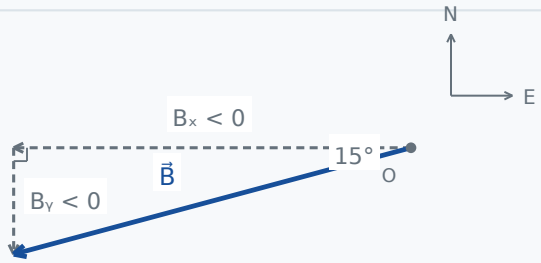
COMPONENT METHOD FOR $\vec{A} + \vec{B}$

3. Add $\vec{A} + \vec{B}$ by components. Draw diagrams of the component legs and the resultant.

$\vec{A}$: east and north components



$\vec{B}$: west and south components



1 Add signed x components and signed y components

$R_x = 4.5 \cos 35° - 3.0 \cos 15° \approx +0.788407$ m

$R_Y = 4.5 \sin 35° - 3.0 \sin 15° \approx +1.804637$ m

Both signs are positive: the result points northeast. The approximate values shown above are checkpoints; use the full expressions below for the final result.

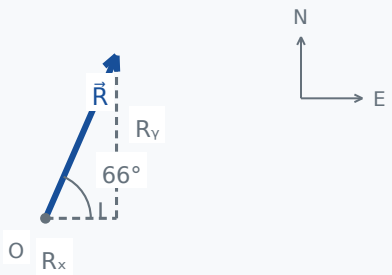
2 Build the resultant triangle

$|\vec{R}| = \sqrt{(R_x^2 + R_Y^2)} \approx 1.969340$ m

$\theta = \tan^{-1}(R_Y / R_x) \approx 66.401°$

Measure θ from east toward north, because the horizontal leg points east.

Resultant and perpendicular legs

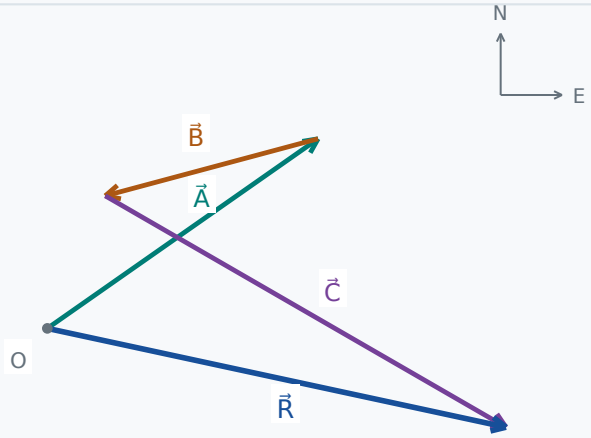


$\vec{A} + \vec{B} = 2.0$ m [E66°N]

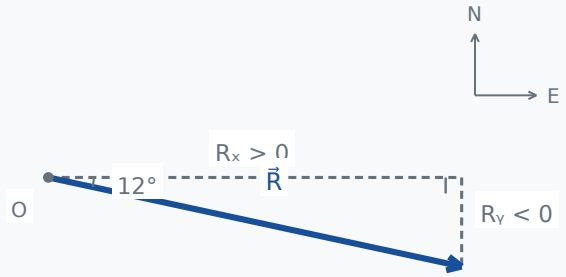
COMPONENT METHOD FOR $\vec{A} + \vec{B} + \vec{C}$

3. Add $\vec{A} + \vec{B} + \vec{C}$ by components; show both the construction and the component triangle.

Three arrows, tip to tail



Resultant from signed x and y totals



1 Add the east-west and north-south columns

$R_x = 4.5 \cos 35° - 3.0 \cos 15° + 6.3 \sin 60°$

$\approx +6.244367$ m

$R_y = 4.5 \sin 35° - 3.0 \sin 15° - 6.3 \cos 60°$

≈ -1.345363 m

2 Find length and choose southeast direction

$|\vec{R}| = \sqrt{(R_x^2 + R_y^2)} \approx 6.387654$ m

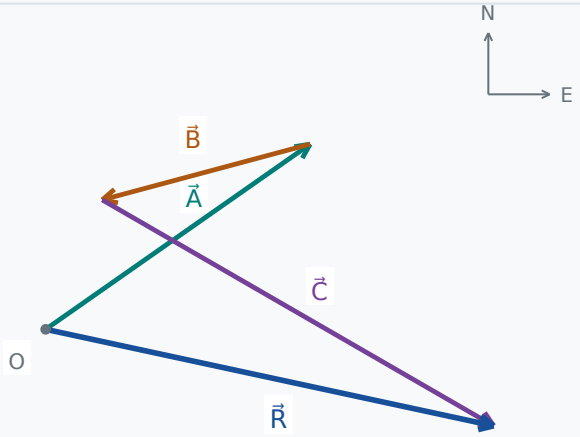
$\theta = \tan^{-1}(|R_y| / R_x) \approx 12.159°$ south of east

$\vec{A} + \vec{B} + \vec{C} = 6.4$ m [E12°S]

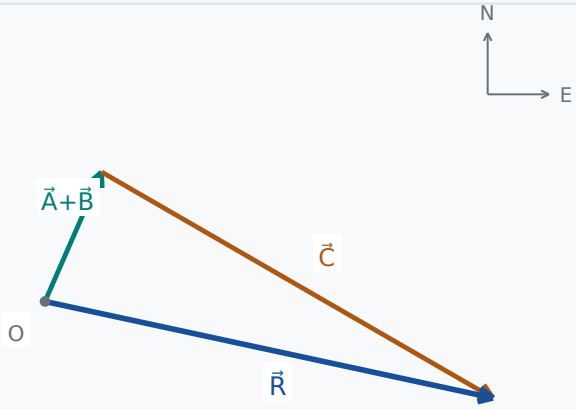
WHY THREE VECTORS NEED TWO TRIANGLES

4. Why were the sine and cosine laws not used directly for $\vec{A} + \vec{B} + \vec{C}$?

Three arrows and a closing resultant



First combine $\vec{A} + \vec{B}$, then add $\vec{C}$



1 **Look at the shape on the left**

Three input arrows plus the blue closing arrow form a four-sided path. Each sine or cosine law relates the sides and angles of one triangle, so it cannot be applied directly to this entire shape.

2 **See how a two-triangle solution would work**

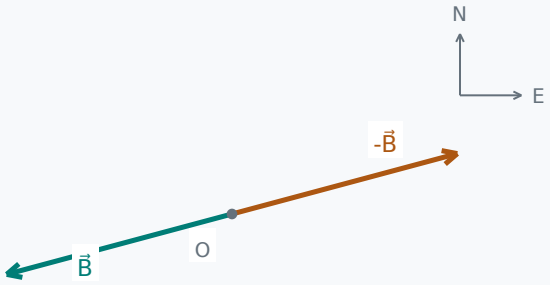
First solve $\vec{A} + \vec{B}$, including both its length and direction. Treat that resultant as a new side; add $\vec{C}$ in a second triangle. Without the direction of $\vec{A} + \vec{B}$, the second included angle is unknown.

Component addition handles all three vectors in one calculation.

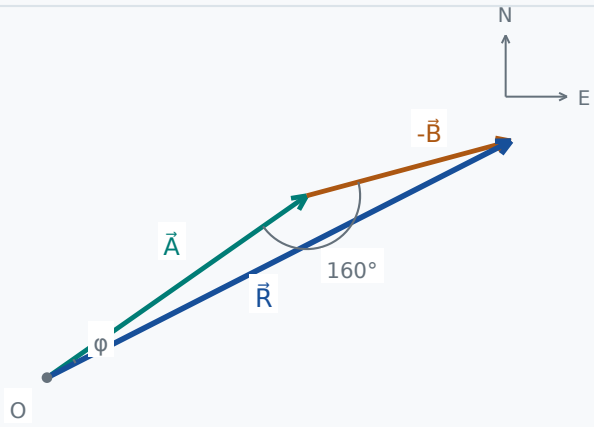
TRIANGLE METHOD FOR $\vec{A}$ - $\vec{B}$

1. Evaluate $\vec{A}$ - $\vec{B}$ by adding the opposite of $\vec{B}$. Draw a diagram and give magnitude and direction.

First reverse $\vec{B}$



Then place -$\vec{B}$ at $\vec{A}$'s tip



1 Find the triangle angle after reversing $\vec{B}$

-$\vec{B}$ is 3.0 m [E15°N]. The directions $\vec{A}$ [E35°N] and -$\vec{B}$ differ by 20°, so the interior angle at $\vec{A}$'s tip is 180° - 20° = 160°.

2 Use the cosine law to find the resultant length

$|\vec{R}|^2 = 4.5^2 + 3.0^2 - 2(4.5)(3.0) \cos 160°$

$|\vec{R}| \approx 7.390650$ m

3 Use the sine law for the direction

$\sin \varphi / 3.0 = \sin 160° / |\vec{R}| \rightarrow \varphi \approx 7.980°$

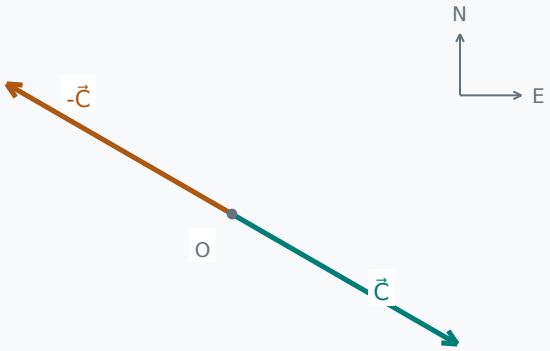
Bearing: [E(35° - φ)N] ≈ [E27.020°N]

$\vec{A}$ - $\vec{B}$ = 7.4 m [E27°N]

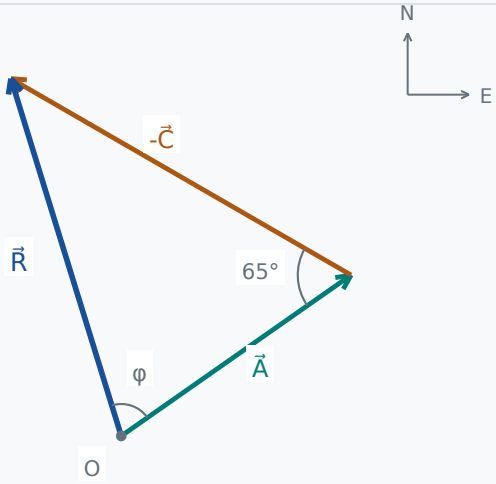
TRIANGLE METHOD FOR $\vec{A}$ - $\vec{C}$

1. Evaluate $\vec{A}$ - $\vec{C}$ by adding the opposite of $\vec{C}$. Draw a diagram and give magnitude and direction.

First reverse $\vec{C}$



Then place -$\vec{C}$ at $\vec{A}$'s tip



1 Find the included angle in the triangle

-$\vec{C}$ is 6.3 m [W30°N], or 150° counterclockwise from east. The backward direction along $\vec{A}$ is 215°, so the interior angle is 215° - 150° = 65°.

2 Use the cosine law for the length

$|\vec{R}|^2 = 4.5^2 + 6.3^2 - 2(4.5)(6.3)\cos 65°$

$|\vec{R}| \approx 5.998128$ m

3 Find φ, then convert it to north of west

$\cos \varphi = (4.5^2 + |\vec{R}|^2 - 6.3^2) / (2(4.5)|\vec{R}|)$

$\varphi \approx 72.161°$; west-to-resultant = 180° - 35° - φ

$\approx 72.839°$

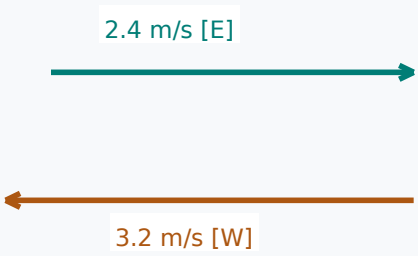
$\vec{A}$ - $\vec{C}$ = 6.0 m [W73°N]

PRACTICE (a): OPPOSING VELOCITIES

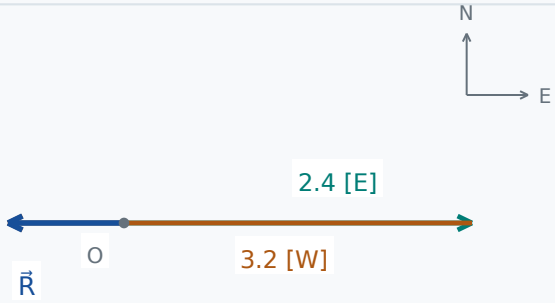
a) 2.4 m/s [E] + 3.2 m/s [W]

1 Draw the two motions on an east-west line

Opposing inputs



Tip-to-tail construction



2 Choose east as positive and add signed values

$R_x = +2.4 - 3.2 = -0.8$ m/s

$R_y = 0$. Negative x means west.

3 Report size and direction with justified precision

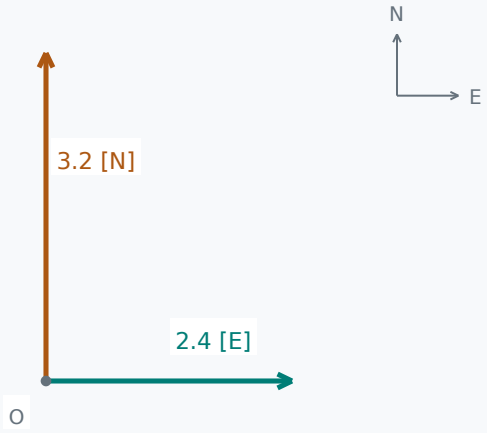
The size is |-0.8| = 0.8 m/s; attach [W] for the direction. The inputs are stated to the nearest 0.1 m/s, so the difference is also known to 0.1 m/s. Writing 0.80 would imply unsupported precision.

a) $\vec{R}$ = 0.8 m/s [W]

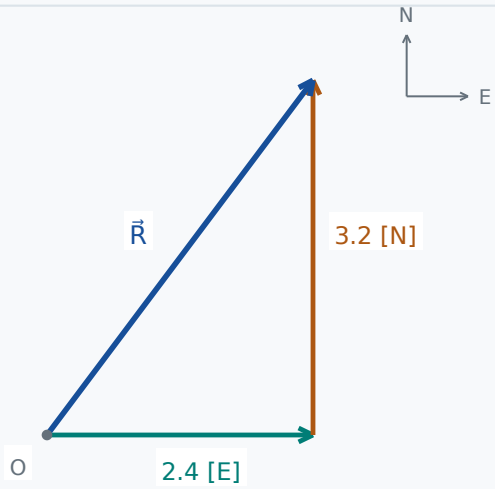
PRACTICE (b): PERPENDICULAR VELOCITIES

b) 2.4 m/s [E] + 3.2 m/s [N]

Given east and north arrows



Place arrows tip to tail



1 Recognize the right triangle

The 2.4 east and 3.2 north arrows are perpendicular legs. The resultant runs from the first tail to the last tip and is the hypotenuse.

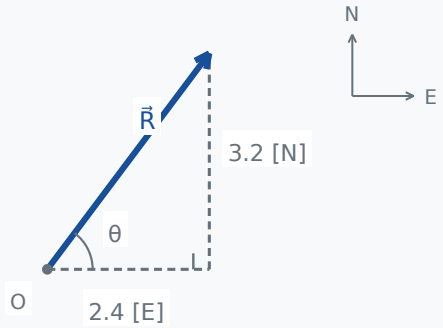
2 Use Pythagoras and tangent

$|\vec{R}| = \sqrt{(2.4^2 + 3.2^2)} = 4.0$ m/s

$\theta = \tan^{-1}(3.2 / 2.4) \approx 53.130°$

Both legs are positive. Measure θ from east toward north.

The same triangle, with angle θ

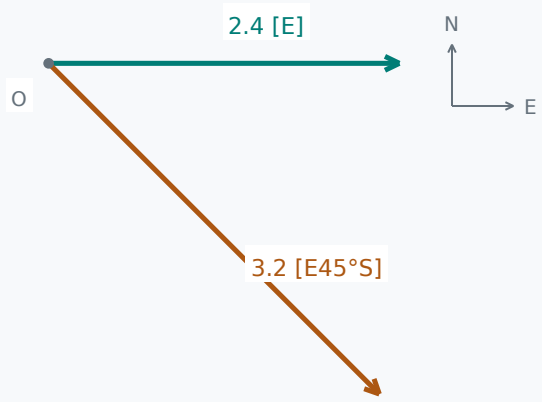


b) $\vec{R}$ = 4.0 m/s [E53°N]

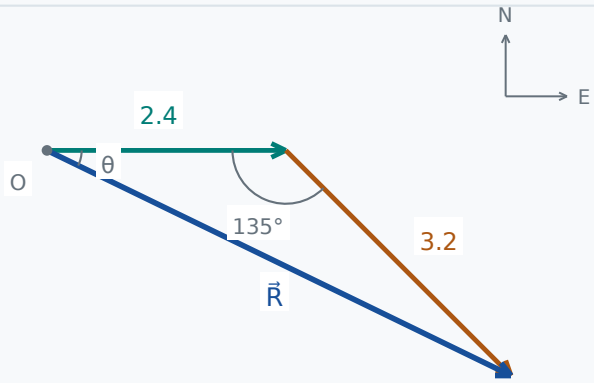
PRACTICE (c): TRIANGLE LAWS

c) 2.4 m/s [E] + 3.2 m/s [E45°S]

Given bearings from one origin



Tip-to-tail triangle



1 Find the interior triangle angle

The given velocity directions differ by 45°. At the join, the first arrow points backward, so the included triangle angle is 180° - 45° = 135°.

2 Use cosine for length, then sine for direction

$|\vec{R}|^2 = 2.4^2 + 3.2^2 - 2(2.4)(3.2)\cos 135°$

$|\vec{R}| \approx 5.182775$ m/s

$\sin\theta / 3.2 = \sin 135° / |\vec{R}|$

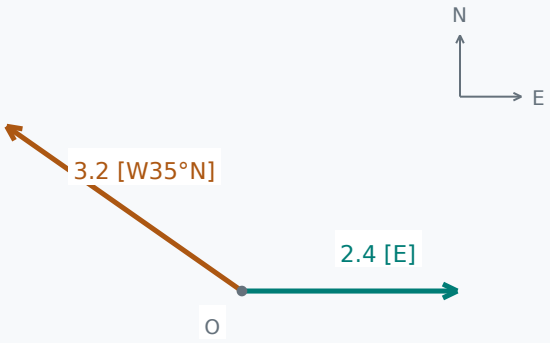
$\theta \approx 25.886°$ below east

c) $\vec{R}$ = 5.2 m/s [E26°S]

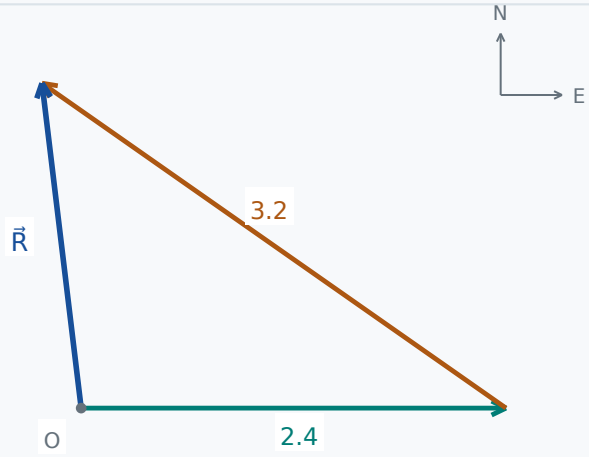
PRACTICE (d): MIXED DIRECTIONS

d) 2.4 m/s [E] + 3.2 m/s [W35°N]

Read the two original bearings



Build the tip-to-tail path



1 Resolve the tilted vector with signed components

The 35° is measured from west: its horizontal part uses cosine and is negative; its vertical part uses sine and is positive.

2 Add signed components

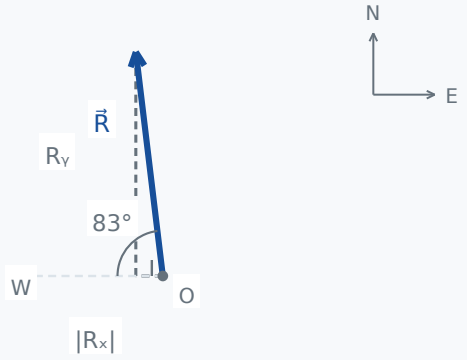
$R_x = 2.4 - 3.2 \cos 35° \approx -0.221287$ m/s

$R_Y = 3.2 \sin 35° \approx +1.835445$ m/s

$|\vec{R}| = \sqrt{(R_x^2 + R_Y^2)} \approx 1.848736$ m/s

$\theta = \tan^{-1}(|R_Y| / |R_x|) \approx 83.125°$

Final right triangle: west and north

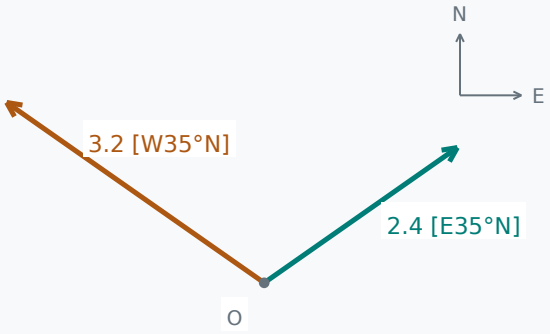


d) $\vec{R}$ = 1.8 m/s [W83°N]

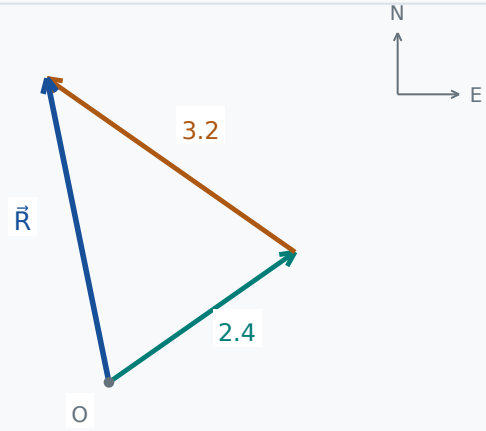
PRACTICE (e): WHY NORTH OF WEST?

e) 2.4 m/s [E35°N] + 3.2 m/s [W35°N]

Both input arrows point partly north



Add them tip to tail



1 Determine the signs before calculating the angle

The west part from 3.2 is larger than the east part from 2.4, so net x is negative. Both north parts add, so net y is positive. The direction must be northwest.

2 Add components and draw the resultant

$R_x = (2.4 - 3.2) \cos 35° \approx -0.655322$ m/s

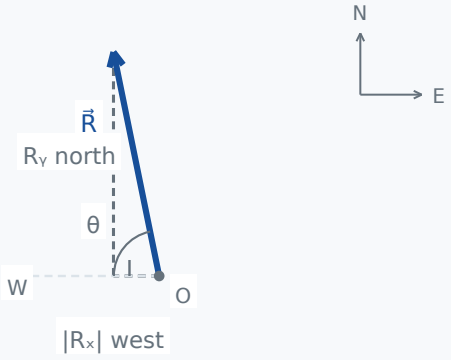
$R_y = (2.4 + 3.2) \sin 35° \approx +3.212028$ m/s

$|\vec{R}| = \sqrt{(R_x^2 + R_y^2)} \approx 3.278196$ m/s

$\theta = \tan^{-1}(|R_y| / |R_x|) \approx 78.469°$

θ begins at west and turns north to the resultant.

The resultant is the hypotenuse

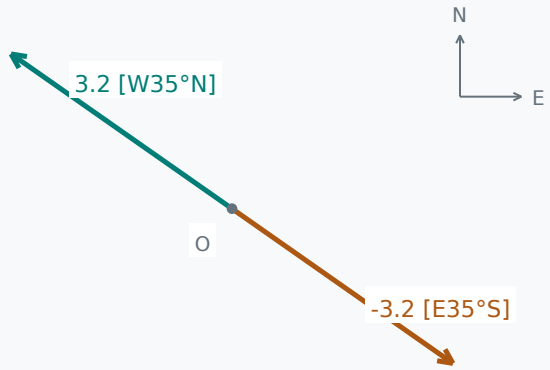


e) $\vec{R}$ = 3.3 m/s [W78°N]

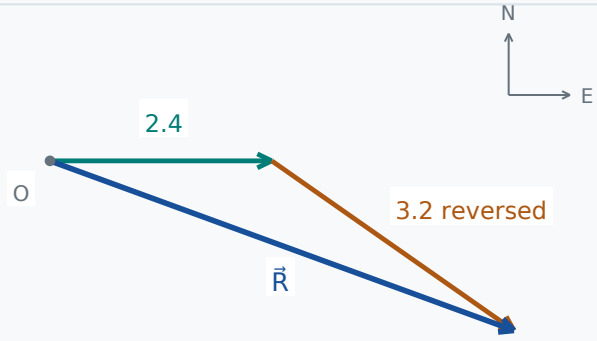
PRACTICE (f): SUBTRACTION BY REVERSAL

f) 2.4 m/s [E] - 3.2 m/s [W35°N]

Opposite directions of the 3.2 vector



Add 2.4 [E] to the reversed arrow



1 Reverse the subtracted velocity

[W35°N] becomes [E35°S]. Reversal changes direction by 180°, but the 3.2 m/s magnitude stays the same.

2 Add signed components

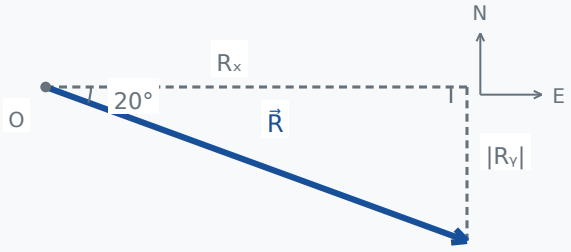
$R_x = 2.4 + 3.2 \cos 35° \approx +5.021287$ m/s

$R_y = -3.2 \sin 35° \approx -1.835445$ m/s

$|\vec{R}| = \sqrt{(R_x^2 + R_y^2)} \approx 5.346230$ m/s

$\theta = \tan^{-1}(|R_y| / R_x) \approx 20.079°$

Final right triangle: east and south



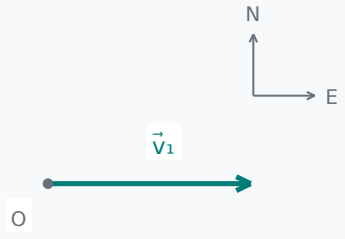
f) $\vec{R}$ = 5.3 m/s [E20°S]

PRACTICE (g): RESOLVE ALL THREE INPUTS

g) 2.4 m/s [E] + 3.2 m/s [W35°N] + 4.2 m/s [S10°W]

1 Draw each vector and mark its signed component legs

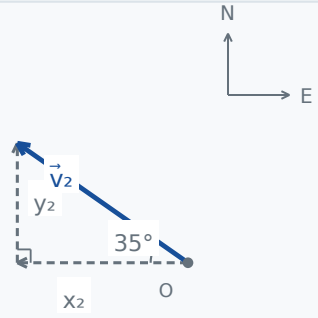
First: 2.4 [E]



$x_1 = +2.4$ m/s

$y_1 = 0$

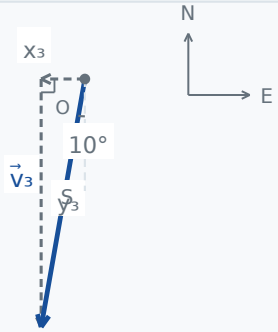
Second: 3.2 [W35°N]



$x_2 = -3.2 \cos 35°$

$y_2 = +3.2 \sin 35°$

Third: 4.2 [S10°W]



$x_3 = -4.2 \sin 10°$

$y_3 = -4.2 \cos 10°$

2 Check the signs against the compass directions

East is +x and north is +y. The second vector points west and north. For [S10°W], the angle begins at south, so its small west part uses sine and its larger south part uses cosine; both are negative.

Signed component summary

x: +2.4, -3.2 cos 35°, -4.2 sin 10° (m/s)

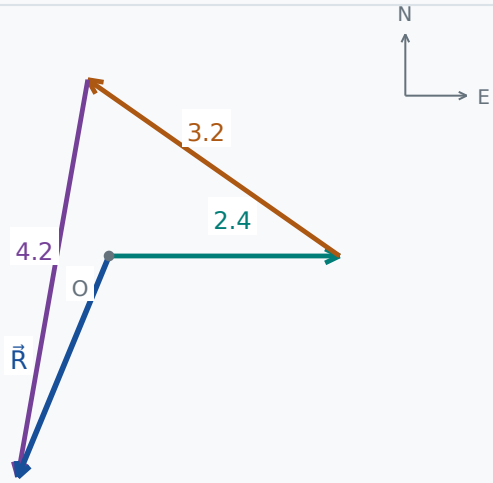
y: 0, +3.2 sin 35°, -4.2 cos 10° (m/s)

Next page: add the two columns and turn the signed totals into magnitude and direction.

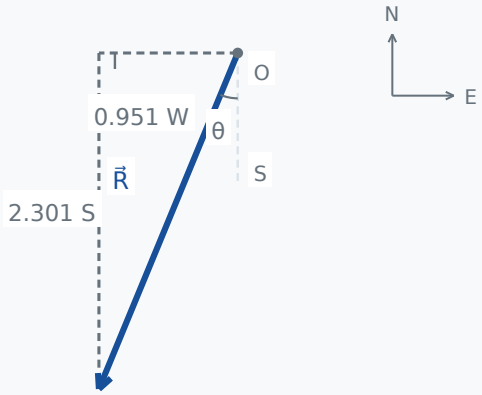
PRACTICE (g): RESULTANT AND DIRECTION

g) Continue: combine the three signed component pairs and state the final resultant.

All three arrows tip to tail



Net component right triangle



1 Sum the signed x and y components

$R_x = 2.4 - 3.2 \cos 35° - 4.2 \sin 10°$

≈ -0.950609 m/s

$R_y = 3.2 \sin 35° - 4.2 \cos 10°$

≈ -2.300748 m/s

2 Use the signs to select the southwest quadrant

$|\vec{R}| = \sqrt{(R_x^2 + R_y^2)} \approx 2.489397$ m/s

$\theta = \tan^{-1}(|R_x| / |R_y|) \approx 22.449°$ west of south

West is opposite θ; south is adjacent. Measure θ from south toward west.

g) $\vec{R}$ = 2.5 m/s [S22°W]

Optional online vector Gizmo: an exploratory activity; no fixed numerical answer.